



Proof that every prime has a primitive root.

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So I encountered this proof on a Number Theory book, I will link the pdf at the end of the post (proof at page 96), it says: **"Every prime has a primitive root, proof: Let p be a prime and let m be a positive integer such that: $p-1=mk$ for some integer k . Let $F(m)$ be the number of positive integers of order m modulo p that are less than p . The order modulo p of an integer not divisible by p divides $p-1$, it follows that:**

$$p-1 = \sum_{m|p-1} F(m)$$

By theorem 42 we know that:

$$p-1 = \sum_{m|p-1} \phi(m)$$

By Lemma 11, $F(m) \leq \phi(m)$ when $m|(p-1)$. Together with:

$$\sum_{m|p-1} F(m) = \sum_{m|p-1} \phi(m)$$

we see that $F(m) = \phi(m)$ for each positive divisor m of $p-1$. Thus we conclude that $F(m) = \phi(m)$. As a result, we see that there are $p-1$ incongruent integers of order $p-1$ modulo p . Thus p has $\phi(p-1)$ primitive roots.

The part that i don't understand is near the beginning, when he says "The order modulo p of an integer not divisible by p divides $p-1$, it follows that:

$$p-1 = \sum_{m|p-1} F(m)$$

" How does he conclude that? I understand that the order of the integer must divide $p-1$ but how does that imply that the summation actually evaluates at $p-1$?

Link of the book's pdf: <https://www.saylor.org/site/wp-content/uploads/2013/05/An-Introductory-in-Elementary-Number-Theory.pdf>

[elementary-number-theory](#)

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edited Apr 8, 2021 at 14:41



Bill Dubuque

272k 38 300 947

asked Jul 23, 2018 at 13:51



Spasoje Durovic

977 7 17

- ▲ Try this for some small prime like 5 or 7 and you'll see. – [saulspatz](#) Jul 23, 2018 at 13:57
- ▲ The nonzero elements of the p -element field form a group under multiplication of order $p - 1$; thus the multiplicative order must be a divisor of $p - 1$. – [egreg](#) Jul 23, 2018 at 14:10
- ▲ Recall how you defined $F(m)$, "Let $F(m)$ be the number of positive integers of order $m \bmod p$ that are less than p ." What's the order of the group? – [sharding4](#) Jul 23, 2018 at 14:29
- ▲ Nice proof actually. Another proof is a corollary of the (much longer) result that if $(F, +, 0, \times, 1)$ is a finite field then the group $(F \setminus \{0\}, \times, 1)$ is cyclic. – [DanielWainfleet](#) Jul 23, 2018 at 15:03

2 Answers

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- There are $p - 1$ positive integers less than p , namely $1, 2, \dots, p - 1$.
- Each of these will have some multiplicative order modulo p . So if we count all those of order 1, all those of order 2, all those of order 3, etc then the total count is $p - 1$. There are $F(1)$ of order 1, $F(2)$ of order 2, etc, so:

$$p - 1 = \sum_{m=1}^{\infty} F(m)$$

However, we know their orders will divide $p - 1$, so almost all the terms in this sum will be zero. Only those with $m | (p - 1)$ will contribute to the sum. We therefore have:

$$p - 1 = \sum_{m|p-1} F(m)$$

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answered Jul 23, 2018 at 14:48

**Jaap Scherphuis**

13.7k 1 19 31

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- All that is saying is that the sum of the number of elements of each order must add up to the total number of elements in the field, $p - 1$. The summation could have been carried out of "all possible orders," but since "all possible orders" is the same thing as "divisors of $p - 1$," we can carry the summation over divisors of $p - 1$. The summation equation is literally equivalent to the statement that the order of a member of the field must be a factor of $p - 1$

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answered Oct 14, 2022 at 5:34

**Tom Healy**

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